

ALS–Tucker Approximation for 2D Function Representation in a Chebyshev Basis

Overview

We approximate a smooth bivariate function

$$f(x, y) \approx (T_x A) G (T_y B)^T,$$

where T_x, T_y are Chebyshev–Vandermonde matrices, $A \in \mathbb{R}^{(d_x+1) \times R}$ and $B \in \mathbb{R}^{(d_y+1) \times R}$ are factor matrices, and $G \in \mathbb{R}^{R \times R}$ is a Tucker core.

The goal of the Alternating Least Squares (ALS) scheme is to minimize

$$\min_{A, B, G} \|F - T_x A G (T_y B)^T\|_F^2,$$

where F contains samples of $f(x, y)$ on a tensor grid.

Metric Matrices

Define the Chebyshev metric matrices

$$G_x := T_x^T T_x, \quad G_y := T_y^T T_y.$$

These store the discrete Chebyshev inner product in coefficient space.

G Update (Closed Form)

For fixed A and B , the minimizer of

$$\|F - T_x A G (T_y B)^T\|_F^2$$

is obtained by setting the gradient w.r.t. G to zero:

$$(T_x A)^T (T_x A) G (T_y B)^T (T_y B) = (T_x A)^T F (T_y B).$$

In metric form:

$$(A^T G_x A) G (B^T G_y B) = A^T T_x^T F T_y B.$$

Thus

$$G = (A^T G_x A)^{-1} (A^T T_x^T F T_y B) (B^T G_y B)^{-1}.$$

A Update (Normal Equation)

With B and G fixed, A solves

$$\min_A \|F - T_x A G (T_y B)^T\|_F^2.$$

The corresponding normal equation is

$$G_x A \underbrace{(G B^T G_y B G^T)}_{\text{“middleA”}} = T_x^T F T_y B G^T.$$

Thus

$$A = G_x^{-1} (T_x^T F T_y B G^T) (G B^T G_y B G^T)^\dagger.$$

B Update (Normal Equation)

Similarly,

$$G_y B \underbrace{(G^T A^T G_x A G)}_{\text{“middleB”}} = T_y^T F^T T_x A G.$$

Thus

$$B = G_y^{-1} (T_y^T F^T T_x A G) (G^T A^T G_x A G)^\dagger.$$

Metric-Orthonormalization

After each update, the columns of A and B are orthonormalized in the Chebyshev metric:

$$A \leftarrow \text{QR in metric } G_x \implies A^T G_x A = I,$$

$$B \leftarrow \text{QR in metric } G_y \implies B^T G_y B = I.$$

Conditioning

Because the orthonormalization enforces

$$A^T G_x A = I, \quad B^T G_y B = I,$$

the matrices appearing in the G update are perfectly conditioned.

However, the matrices in the normal equations contain

$$\text{middleA} = G(B^T G_y B)G^T \approx GG^T,$$

$$\text{middleB} = G^T(A^T G_x A)G \approx G^T G.$$

Thus their condition numbers satisfy

$$\kappa(\text{middleA}) = \kappa(GG^T) = \kappa(G)^2,$$

$$\kappa(\text{middleB}) = \kappa(G^T G) = \kappa(G)^2.$$

I think here: If ALS forces extremely small singular values into G (typical when the requested rank exceeds the effective rank of f), then G becomes nearly rank-deficient and

$$\kappa(GG^T), \kappa(G^T G) \gg 1,$$

even though $A^T G_x A$ and $B^T G_y B$ remain perfectly conditioned.

This probably explains why the algorithm exhibits large condition numbers in the middle matrices despite exact metric orthonormality.

Typical numerical example. A typical run with a high rank produces values of this form:

$$\text{cond}(\text{middle_A}) = 4.171459120928041 \times 10^{31},$$

$$\text{cond}(\text{middle_B}) = 4.188845152793778 \times 10^{31},$$

$$\text{cond}(A^T G_x A) = 1.0000000000000004,$$

$$\text{cond}(B^T G_y B) = 1.0000000000000009.$$